

Best-Sweep 1D Learning Pipeline Updates and 10-Replica Results

NN_FPN repository notes

June 12, 2026

Learning-pipeline changes relative to the paper reproduction setup

The experiments here keep the paper evaluation metric but change the learning pipeline used to train the filter models. The reported quantity is the scalar-flux error ratio

$$r_N = \frac{\|\phi^{FPN} - \phi^{\text{exact}}\|}{\|\phi^{P_N} - \phi^{\text{exact}}\|},$$

so smaller values are better. The neural-network and constant-filter entries are denoted by r_N^{nn} and r_N^{const} , respectively. The main changes are:

- Training uses the augmented 1D sampler instead of the original fixed four-family paper-mode sampler. Each batch draws randomized initial/source families and randomized material families, including Gaussian, discontinuous, vanishing-cross-section-like, source-driven, and Reeds-like cases.
- The NN architecture and optimizer settings are selected from W&B random sweeps minimizing **train/loss**; the winning settings are then retrained for 10 independent replicas.
- The feature map is sweep-selected per moment order. In particular, $N = 3$ uses **log_material_only**; $N = 7$ uses **no_norm_log**; and $N = 9$ uses **baseline_norm**. All winning settings train on final-time scalar-flux loss with no auxiliary moment penalty.
- Constant-filter models are retrained on the same selected training settings and replica count for the corresponding N , then evaluated against the same tests.

N	feature variant	log scale	log clip	material normalization	material channels
3	log_material_only	0.1	[0, 20]	none	scale logs + ratios
7	no_norm_log	0.1	[0, 10]	sample	scale logs + ratios
9	baseline_norm	1.0	[0, 40]	sample	scale logs

Selected learning hyperparameters

All rows use the augmented training sampler, final-time scalar-flux loss, **aux_moment_loss_weight** = 0, and **relative_loss_eps** = 10^{-12} .

N	replicas	hidden	batch	epochs	optimizer	lr	scheduler (warmup, min)	wd NN / const	horizons
3	10	64	32	100	Adam	0.003	constant (0, 10^{-5})	$10^{-3}/10^{-4}$	{0.5}
7	10	512	64	100	Adam	0.01	cosine (0.05, 0)	$10^{-6}/10^{-4}$	{0.5}
9	10	256	32	100	Adam	0.01	constant (0, 0)	$10^{-6}/10^{-4}$	{0.25, 0.5, 1.0}

Neural-network input features

For a training sample and spatial cell i , let $u_i = (u_{i,0}, \dots, u_{i,N})^\top \in \mathbb{R}^{N+1}$ denote the moment vector from the previous iterate. The transport residual used by the code is

$$d_i = \frac{(Au)_{i+1/2} - (Au)_{i-1/2}}{\Delta x},$$

where the interface quantities are the upwind finite-volume fluxes assembled in the solver. The remaining baseline quantities are

$$c_i = \sigma_{t,i} u_i, \quad s_i = \sigma_{s,i} u_{i,0}, \quad q_i = Q_i,$$

with $d_i, c_i \in \mathbb{R}^{N+1}$ and $s_i, q_i \in \mathbb{R}$. For any channel tensor $g_{i,k}$, the sample-wise spatial normalization is

$$\mathcal{N}_x(g)_{i,k} = \frac{g_{i,k} - \frac{1}{n_x} \sum_{j=1}^{n_x} g_{j,k}}{\text{std}_{1 \leq j \leq n_x}(g_{j,k}) + 10^{-10}}.$$

With the filter type used in the sweeps, the manuscript-style baseline input is

$$z_i^{\text{base}} = \left[\mathcal{N}_x(|d_i|), \mathcal{N}_x(|c_i|), \mathcal{N}_x(|s_i|), \mathcal{N}_x(|q_i|) \right] \in \mathbb{R}^{2N+4}.$$

The log feature map introduced in the sweeps is

$$L_{\alpha,[a,b]}(g) = \text{clip}_{[a,b]} \left(\log \left(1 + \frac{|g|}{\alpha} \right) \right).$$

When material-scale features are enabled, the absorption coefficient is $\sigma_{a,i} = \max(\sigma_{t,i} - \sigma_{s,i}, 0)$ and the appended material vector is

$$m_i = \left[L_{\alpha,[a,b]}(\sigma_{s,i}), L_{\alpha,[a,b]}(\sigma_{t,i}), L_{\alpha,[a,b]}(\sigma_{a,i}), \frac{\sigma_{s,i}}{\max(|\sigma_{t,i}|, \varepsilon)}, \frac{\sigma_{a,i}}{\max(|\sigma_{t,i}|, \varepsilon)} \right].$$

If material ratios are disabled, only the first three scale-log entries of m_i are appended. Thus the three reported best configurations use

$$\begin{aligned} x_i^{N=3} &= [z_i^{\text{base}}, m_i], & \alpha &= 0.1, \quad [a, b] = [0, 20], \quad \dim x_i = 15, \\ x_i^{N=7} &= [L_{0.1,[0,10]}(d_i), L_{0.1,[0,10]}(c_i), L_{0.1,[0,10]}(s_i), L_{0.1,[0,10]}(q_i), \mathcal{N}_x(m_i)], & [a, b] &= [0, 10], \quad \dim x_i = 23, \\ x_i^{N=9} &= [z_i^{\text{base}}, \mathcal{N}_x(m_i^{1:3})], & \alpha &= 1.0, \quad [a, b] = [0, 40], \quad \dim x_i = 25. \end{aligned}$$

The neural network evaluates one scalar per cell, $\sigma_{f,i} = F_\theta(x_i)$, which is used as the learned filter strength in the moment update.

All tables below report mean $\pm$ standard deviation over 10 trained replicas. Boldface marks the lower mean between the NN and constant filter at fixed test case, time, and moment order.

Results tables

Gaussian

t_f	Neural network			Constant filter		
	r_3^{nn}	r_7^{nn}	r_9^{nn}	r_3^{const}	r_7^{const}	r_9^{const}
0.5	0.2467 $\pm$ 0.0098	0.1089 $\pm$ 0.0169	0.1108 $\pm$ 0.0788	0.9814 $\pm$ 0.0010	0.9618 $\pm$ 0.0012	0.9543 $\pm$ 0.0042
1.0	0.2096 $\pm$ 0.0023	0.1036 $\pm$ 0.0400	0.0363 $\pm$ 0.0222	0.9647 $\pm$ 0.0018	0.9237 $\pm$ 0.0023	0.9101 $\pm$ 0.0081

Vanishing cross section

t_f	Neural network			Constant filter		
	r_3^{nn}	r_7^{nn}	r_9^{nn}	r_3^{const}	r_7^{const}	r_9^{const}
0.5	0.3339 $\pm$ 0.0091	0.0881 $\pm$ 0.0200	0.1040 $\pm$ 0.0749	0.9835 $\pm$ 0.0009	0.9613 $\pm$ 0.0012	0.9552 $\pm$ 0.0041
1.0	0.0817 $\pm$ 0.0048	0.0896 $\pm$ 0.0401	0.0274 $\pm$ 0.0181	0.9621 $\pm$ 0.0020	0.9263 $\pm$ 0.0022	0.9106 $\pm$ 0.0081

Discontinuous cross section

t_f	Neural network			Constant filter		
	r_3^{nn}	r_7^{nn}	r_9^{nn}	r_3^{const}	r_7^{const}	r_9^{const}
0.5	0.3151 $\pm$ 0.0083	0.1050 $\pm$ 0.0208	0.1093 $\pm$ 0.0977	0.9829 $\pm$ 0.0009	0.9607 $\pm$ 0.0012	0.9547 $\pm$ 0.0042
1.0	0.1991 $\pm$ 0.0088	0.1479 $\pm$ 0.0382	0.0480 $\pm$ 0.0227	0.9623 $\pm$ 0.0020	0.9242 $\pm$ 0.0023	0.9094 $\pm$ 0.0082

Reeds

t_f	Neural network			Constant filter		
	r_3^{nn}	r_7^{nn}	r_9^{nn}	r_3^{const}	r_7^{const}	r_9^{const}
5.0	0.4493 ± 0.0099	0.4725 ± 0.0813	0.3582 ± 0.0330	0.8638 ± 0.0066	0.7413 ± 0.0067	0.7037 ± 0.0225
10.0	0.7526 ± 0.0128	0.8206 ± 0.1134	0.5745 ± 0.0728	0.8215 ± 0.0077	0.5277 ± 0.0093	0.4807 ± 0.0296

Corresponding figures

Each panel shows the saved scalar-flux/filter plot from the corresponding 10-replica run directory. For each test case and time, the columns are NN $N = 3$, constant $N = 3$, NN $N = 7$, constant $N = 7$, NN $N = 9$, and constant $N = 9$.

Gaussian

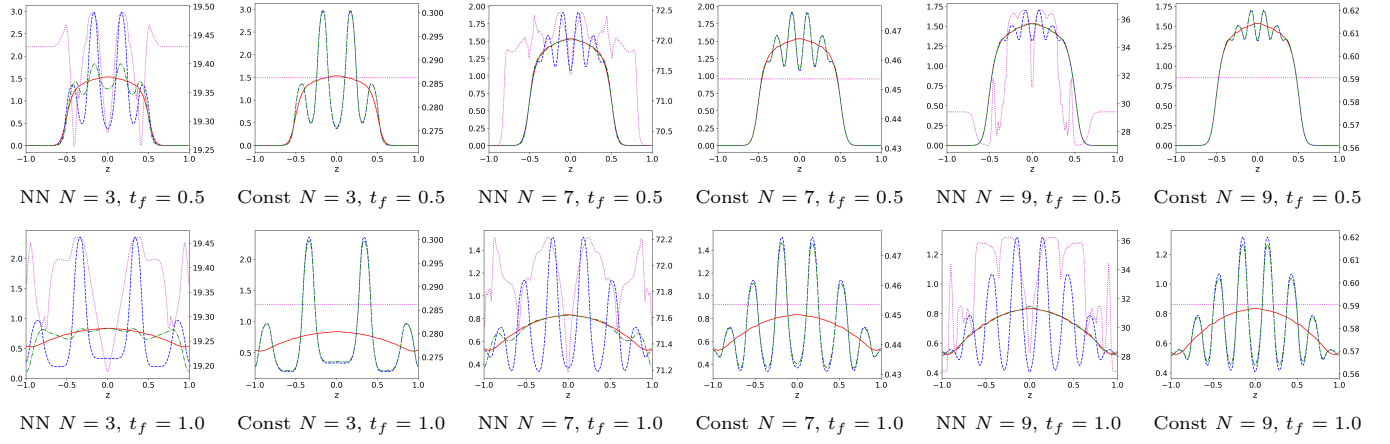


Figure 1: Gaussian test case.

Vanishing cross section

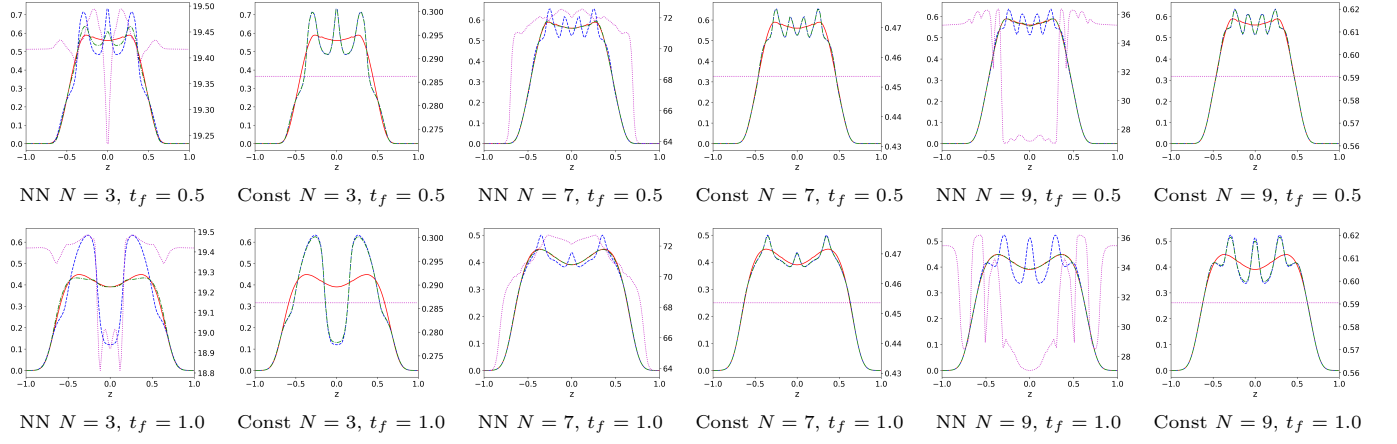


Figure 2: Vanishing cross section test case.

Discontinuous cross section

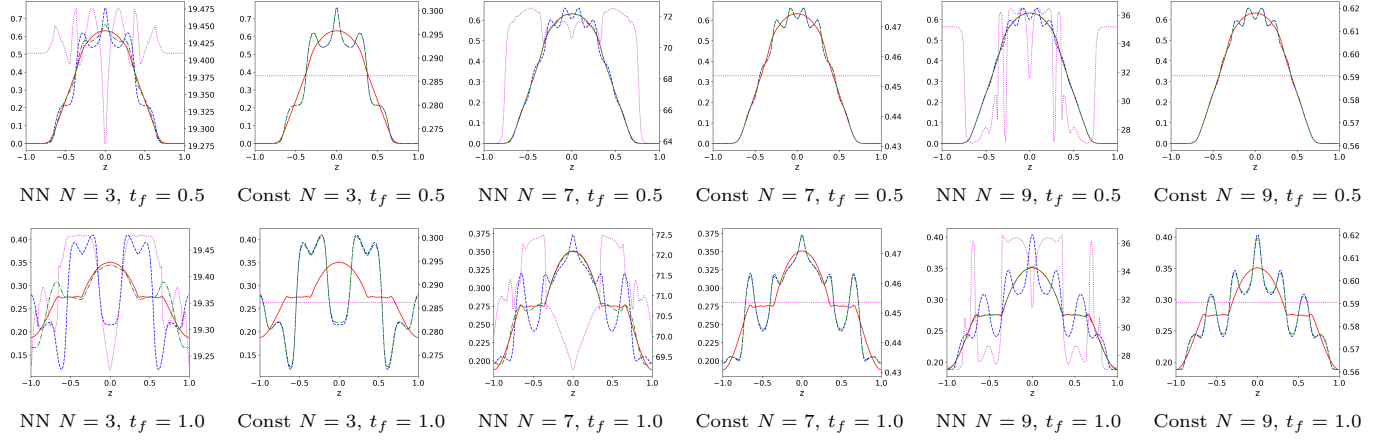


Figure 3: Discontinuous cross section test case.

Reeds

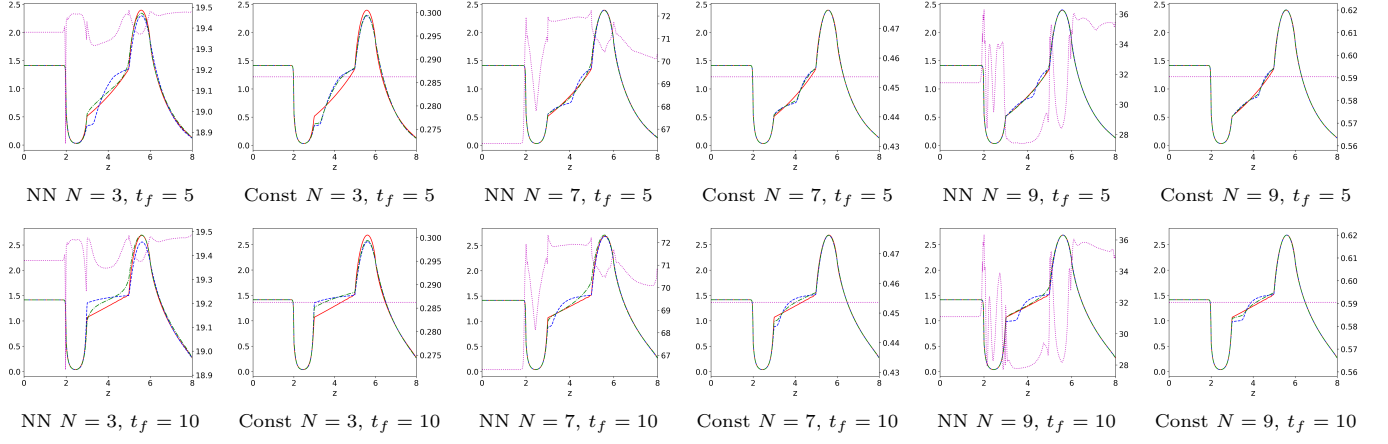


Figure 4: Reeds test case.

Paper-setting constant-filter baseline

As a third baseline, the constant-filter ansatz was trained with the original paper-style 1D training sampler rather than the augmented sweep-selected sampler. The reported values are the same scalar-flux error ratio r_N , averaged over 10 independently trained constant-filter replicas.

Learning hyperparameters

N	ansatz	replicas	training data	batch	epochs	lr schedule	weight decay	horizons
3	constant	10	paper	64	200	cosine, lr 0.1, warmup 0.05	10^{-6}	{0.25, 0.5, 1.0}
7	constant	10	paper	64	200	cosine, lr 0.1, warmup 0.05	10^{-6}	{0.25, 0.5, 1.0}
9	constant	10	paper	64	200	cosine, lr 0.1, warmup 0.05	10^{-6}	{0.25, 0.5, 1.0}

All rows use Adam, `num_IC` = 4, `sigs_max` = 1, final-time scalar-flux loss, `aux_moment_loss_weight` = 0, and model tag `paper_constant_baseline`.

Evaluation results

Problem	t_f	r_3^{const}	r_7^{const}	r_9^{const}
Gaussian	0.5	0.5772 ± 0.0009	0.5325 ± 0.0012	0.5344 ± 0.0013
Gaussian	1.0	0.3636 ± 0.0010	0.2790 ± 0.0013	0.2793 ± 0.0014
Vanishing cross section	0.5	0.6267 ± 0.0008	0.5252 ± 0.0012	0.5330 ± 0.0014
Vanishing cross section	1.0	0.2975 ± 0.0012	0.2841 ± 0.0014	0.2734 ± 0.0015
Discontinuous cross section	0.5	0.6114 ± 0.0008	0.5234 ± 0.0012	0.5311 ± 0.0014
Discontinuous cross section	1.0	0.2942 ± 0.0012	0.2772 ± 0.0014	0.2703 ± 0.0015
Reeds	5.0	0.2613 ± 0.0002	0.2502 ± 0.0002	0.2438 ± 0.0002
Reeds	10.0	0.5550 ± 0.0000	0.4007 ± 0.0006	0.3242 ± 0.0005